

The VB program has three main forms with different text box command buttons as explained below. The first form is designed such that the VB program first accesses the database records.

If Show button is pressed followed by Show Record button, Form2 is opened with four menu buttons, namely, Select COM port, Connect Device, Exit and Next (refer Fig. 6). As shown in Form2, user needs to select the item and then connect to the microcontroller through serial COM port of the PC.

The minimum, maximum and grace timings are displayed on the user's PC screen that is being used to monitor the show. These timings are sent to the microcontroller through RS232 communication interface. The communication between PC and microcontroller is established using MSCOMM control in VB.

[illegible]

The screenshot shows a web browser window with the address bar displaying "http://www.efy.com/". The page content includes the text "EFY ENTERPRISES PVT. LTD." and "NEW DELHI" in black. Below this, a red banner reads "LIVE SHOW PROGRAMMABLE TIMER". The main heading is "FOLK SONG GENERAL" in large black letters. There are three digital clock displays: "Main Time: 00:00:00", "Back Time: 00:00:00", and "Reset Time: 00:00:00". A large blue digital clock in the center shows "00:00:00". At the bottom, there are five buttons: "Cancel", "Back", "Stop", "Start", and "Clear".

Firmware. The program for microcontroller was written in Basic and compiled using Proton IDE v2.0. The hex code generated by Proton IDE is used to burn into the PIC16F877A using PICSTART Plus programmer.

An actual-size, single-side PCB layout of the circuit is shown in Fig. 8 and its component layout in Fig. 9. All the components are easily available in electronics components shops. After assembling the components on the PCB, you can connect the LCD and three bulbs. Ensure that power supply connections are properly checked.

program section. Select the exact COM port from the drop-down menu near the Select COMPORT box. To know which COM port you are using, you can view it from the Device Manager of your PC. Note that you need to install an appropriate driver for USB-to-serial converter.

is displayed on the LCD module (refer Fig. 10). If this message is not received in the LCD, reboot your system and try it again. Vary VR1 to change the contrast of the LCD. Also check voltage levels at various points in the circuit as listed in the test points table.